UNIVERSIDAD DE PAMPLONA

Demostraciones de la Función y Distribución Beta

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Función Beta

$$\bullet\ B(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1}\,dx \quad , \quad a,b>0$$

Propiedades:

$\star\ B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$

$$\bullet\ \Gamma(a) = \int_0^\infty u^{a-1}e^{-u}\,du$$

$$\bullet\ \Gamma(b) = \int_0^\infty v^{b-1}e^{-v}\,dv$$

$$\Rightarrow \Gamma(a)\Gamma(b) = \int_0^\infty u^{a-1}e^{-u}\,du \int_0^\infty v^{b-1}e^{-v}\,dv$$

$$\Rightarrow \cdots = \int_0^\infty \int_0^\infty \left(u^{a-1}e^{-u}v^{b-1}e^{-v}\right)\,du\,dv$$

$$\begin{aligned} &\star\ u = zt \\ &\star\ v = z(1-t) \end{aligned} \qquad t\in[0,1]$$

$$\star\ du = t\,dz + z\,dt$$

$$\star\ dv = (1-t)\,dz - z\,dt$$

$$\star\ |\det(J)| = \left|\det\begin{pmatrix} \frac{\partial u}{\partial z} & \frac{\partial u}{\partial t} \\ \frac{\partial v}{\partial z} & \frac{\partial v}{\partial t} \end{pmatrix}\right|$$

$$= \left|\det\begin{pmatrix} t & z \\ 1-t & -z \end{pmatrix}\right|$$

$$= |-tz - z(1-t)|$$

$$= |-tz - z + tz|$$

$$= |z|$$

$$= z$$

$$\Rightarrow \Gamma(a)\Gamma(b) = \int_0^1 \int_0^\infty u^{a-1} v^{b-1} e^{-u-v}\, du\, dv$$

$$\Rightarrow \cdots = \int_0^1 \int_0^\infty (zt)^{a-1} [z(1-t)]^{b-1} e^{-zt-z(1-t)} z\, dz\, dt$$

$$\Rightarrow \cdots = \int_0^1 \int_0^\infty z^{a-1} t^{a-1} z^{b-1} (1-t)^{b-1} e^{-z} z\, dz\, dt$$

$$\Rightarrow \cdots = \int_0^1 \int_0^\infty z^{a+b-1+1-1} e^{-z} e^{a-1} (1-t)^{b-1} e^{-z} z\, dz\, dt$$

$$\Rightarrow \cdots = \int_0^\infty z^{a+b-1} e^{-z}\, dz \int_0^1 t^{a-1} (1-t)^{b-1}\, dt$$

$$\Rightarrow \Gamma(a)\Gamma(b) = \Gamma(a+b) B(a,b)$$

$$\boxed{\therefore B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}}$$

$\star\ B(a,b) = B(b,a)$

$$\star\ B(b,a) = \int_0^1 x^{b-1}(1-x)^{a-1}\,dx$$

$\star\ u = 1 - x \Rightarrow x = 1 - u$

$\star\ du = -dx \Rightarrow -du = dx$

- $x = 0 \Rightarrow u = 1$
- $x = 1 \Rightarrow u = 0$

$$\Rightarrow \cdots = \int_1^0 (1-u)^{b-1}u^{a-1}(-du)$$

$$\Rightarrow \cdots = -\int_1^0 u^{a-1}(1-u)^{b-1}du$$

$$\Rightarrow \cdots = \int_0^1 u^{a-1}(1-u)^{b-1}du$$

$$\boxed{\therefore B(b,a) = B(a,b)}$$

$\star\ B(a,1) = \frac{1}{a}$

$$\star\ B(a,1) = \int_0^1 x^{a-1}(1-x)^{1-1}\,dx$$

$$\Rightarrow \cdots = \int_0^1 x^{a-1}\,dx$$

$$\Rightarrow \cdots = \frac{x^a}{a}\Big|_0^1$$

$$\Rightarrow \cdots = \frac{1^a}{a} - \frac{0^a}{a}$$

$$\boxed{\therefore B(a,1) = \frac{1}{a}}$$

$\star\ B(a+1,b) = \frac{a}{a+b}B(a,b)$

$$\star\ B(a+1,b) = \int_0^1 x^{a+1-1}(1-x)^{b-1}\,dx$$

$$\Rightarrow \cdots = \int_0^1 x^a(1-x)^{b-1}\,dx$$

$$\star\ u = x^a \qquad \star\ dv = (1-x)^{b-1}\,dx$$
$$\star\ du = ax^{a-1}\,dx \qquad \star\ v = -\tfrac{(1-x)^b}{b}$$

$$\Rightarrow \cdots = \left[-\frac{x^a(1-x)^b}{b}\right]_0^1 + \int_0^1 \frac{(1-x)^b}{b}ax^{a-1}\,dx$$
$$\Rightarrow \cdots = -\frac{1^a(1-1)^b}{b} - \left[-\frac{0^a(1-0)^b}{b}\right] + \frac{a}{b}\int_0^1 x^{a-1}(1-x)^b\,dx$$

$$\star\ (1-x)^b = (1-x)^{b-1}(1-x) = (1-x)^{b-1} - x(1-x)^{b-1}$$

$$\Rightarrow \cdots = \frac{a}{b}\left[\int_0^1 x^{a-1}(1-x)^{b-1}\,dx - \int_0^1 x^a x^{a-1}x^1(1-x)^{b-1}\,dx\right]$$
$$\Rightarrow \cdots = \frac{a}{b}[B(a,b) - B(a+1,b)]$$
$$\Rightarrow B(a+1,b) = \frac{a}{b}B(a,b) - \frac{a}{b}B(a+1,b)$$
$$\Rightarrow bB(a+1,b) + aB(a+1,b) = aB(a,b)$$
$$\Rightarrow (a+b)B(a+1,b) = aB(a,b)$$

$$\boxed{\therefore B(a+1,b) = \frac{a}{a+b}B(a,b)}$$

Demostración alternativa:

$$\star\ B(a+1,b) = \frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+1+b)}$$

$$\Rightarrow \cdots = \frac{\Gamma(a+1)\Gamma(b)}{\Gamma((a+b)+1)}$$
$$\Rightarrow \cdots = \frac{a\Gamma(a)\Gamma(b)}{(a+b)\Gamma(a+b)}$$
$$\Rightarrow \cdots = \frac{a}{a+b}\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

$$\boxed{\therefore B(a+1,b) = \frac{a}{a+b}B(a,b)}$$

Distribución Beta

$$f_X(x) = \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1} \quad , \quad 0 < x < 1$$

Demostración $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$

$$\begin{aligned}
\star \int_{-\infty}^{\infty} f_X(x)\,dx &= \int_{-\infty}^{0} f_X(x)\,dx + \int_{0}^{1} f_X(x)\,dx + \int_{1}^{\infty} f_X(x)\,dx \\
&\Rightarrow \cdots = \int_0^1 \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}\,dx \\
&\Rightarrow \cdots = \frac{1}{B(a,b)} \int_0^1 x^{a-1}(1-x)^{b-1}\,dx \\
&\Rightarrow \cdots = \frac{1}{B(a,b)} B(a,b)
\end{aligned}$$

$$\boxed{\therefore \int_{-\infty}^{\infty} \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}\,dx = 1}$$

Demostración $E[X]$

$$E[X] = \frac{a}{a+b}$$

$$\begin{aligned}
E[X] &= \int_{-\infty}^{\infty} x f_X(x)\,dx \\
&\Rightarrow \cdots = \int_{-\infty}^{0} x f_X(x)\,dx + \int_{0}^{1} x f_X(x)\,dx + \int_{1}^{\infty} x f_X(x)\,dx \\
&\Rightarrow \cdots = \int_0^1 x \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}\,dx \\
&\Rightarrow \cdots = \frac{1}{B(a,b)} \int_0^1 x^{(a+1)-1}(1-x)^{b-1}\,dx \\
&\Rightarrow \cdots = \frac{1}{B(a,b)} B(a+1,b) \\
&\Rightarrow \cdots = \frac{1}{B(a,b)} \frac{a}{a+b} B(a,b)
\end{aligned}$$

$$\boxed{\therefore E[X] = \frac{a}{a+b}}$$

Demostración $E[X^2]$

$$
\begin{aligned}
E[X^2] &= \int_0^1 x^2 \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}\,dx \\
\Rightarrow \cdots &= \frac{1}{B(a,b)} \int_0^1 x^{(a+1)+1-1}(1-x)^{b-1}\,dx \\
\Rightarrow \cdots &= \frac{1}{B(a,b)} [B((a+1)+1,b)] \\
\Rightarrow \cdots &= \frac{1}{B(a,b)} \left(\frac{a+1}{a+b+1}\right) B(a+1,b) \\
\Rightarrow \cdots &= \frac{1}{B(a,b)} \left(\frac{a+1}{a+b+1}\right) \left(\frac{a}{a+b}\right) B(a,b)
\end{aligned}
$$

$$
\boxed{E[X^2] = \left(\frac{a+1}{a+b+1}\right) \left(\frac{a}{a+b}\right)}
$$

Demostración $\mathrm{Var}[X]$

$$
\begin{aligned}
\mathrm{Var}[X] &= E[X^2] - (E[X])^2 \\
\Rightarrow \cdots &= \left(\frac{a+1}{a+b+1}\right) \left(\frac{a}{a+b}\right) - \frac{a^2}{(a+b)^2} \\
\Rightarrow \cdots &= \frac{a(a+1)}{(a+b)(a+b+1)} - \frac{a^2}{(a+b)^2} \\
\Rightarrow \cdots &= \frac{a(a+1)(a+b) - a^2(a+b+1)}{(a+b)(a+b+1)(a+b)} \\
\Rightarrow \cdots &= \frac{a(a+1)(a+b) - a(a+b)(a+b+1)}{(a+b)(a+b+1)(a+b)} \\
\Rightarrow \cdots &= \frac{a(a^2+ab+a+b-a^2-ab-a)}{(a+b+1)(a+b)^2}
\end{aligned}
$$

$$
\boxed{\therefore \mathrm{Var}[X] = \frac{ab}{(a+b+1)(a+b)^2}}
$$